



Stochastic Modeling of Glaciar Melting via Monte Carlo Simulations.

Course: CSE400 – Fundamentals of Probability in Computing
Domain: Climate
Milestone 3

GROUP-12

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Project Concept & Motivation

Predicting melt rates is difficult because several environmental variables exhibit natural variability.

Key uncertainties arise from:

- Air temperature fluctuations
- Precipitation variability
- Snowfall threshold
- Cloudiness
- Ice albedo changes

Understanding how these uncertainties affect glacier melt.

Project Objective and Deterministic Baseline

Objective

To simulate glacier melt by modeling environmental uncertainty using probabilistic variables and validating their behavior through Monte Carlo simulation.

Deterministic Baseline Strategy

The deterministic baseline uses fixed physical equations governing glacier melt, while environmental inputs are generated from predefined statistical distributions.

This baseline is reasonable because:

Glacier melt physics is deterministic

Environmental inputs vary naturally

Deterministic energy balance models are widely used in glaciology research

System Pipeline Overview

The simulation pipeline models the glacier melt process from environmental inputs.

Pipeline

Environmental Inputs



Random Variable Generation



Energy Balance Model



Surface Melt Calculation

Uncertainty enters through the environmental variables used as inputs to the physical melt equations.

Sources of Uncertainty

Uncertainty in the glacier system arises from multiple environmental factors:

- Measurement Noise
- Sensor inaccuracies in weather stations
- Natural Climate Variability
- Temperature fluctuations
- Variability in precipitation intensity
- Surface Property Changes
- Ice albedo variation
- Cloud cover variability

These uncertainties are modeled using probability distributions for each environmental variable.



Key Random Variables

Five environmental variables were modeled probabilistically:

Random Variables	Distribution Type	Role in Simulation
Precipitation	Gamma	Influences snow accumulation and melt dynamics
Ice albedo	Truncated Normal	Controls solar radiation absorbtion
Snowfall threshold	Normal	Influences glacier mass balance
Cloudiness	Normal	influences surface energy balance and total glacier melt
Air temperature	Normal	Affects melting energy



Probabilistic Modeling with Deterministic Decisions

Mathematical Assumptions:

- Environmental variables follow specific probability distributions.
- Samples are drawn independently across Monte Carlo runs.

How Uncertainty Interacts with Fixed Decision Rule:

- Inputs are random.
- The melt computation model remains deterministic.
- The same equations are applied to every sampled scenario.



Probabilistic Modeling with Deterministic Decisions

Mathematical equations involving the bounds

SNOWFALL THRESHOLD

$$T_s \sim N(1.0, 0.5^2)$$

$$P_{snow}(T) = \Phi \left(\frac{T_s - T}{\sigma_t} \right)$$



Probabilistic Modeling with Deterministic Decisions

Mathematical equations involving the bounds

PRECIPITATION

$P \sim \text{Gamma}(k, \theta)$

$k=2.5, \theta=3$

Precipitation ≥ 0

Precipitation contributes to both snow accumulation and
rain-induced melt processes.



Probabilistic Modeling with Deterministic Decisions

Mathematical equations involving the bounds

AIR TEMPERATURE

$$T_a \sim N(\mu, \sigma^2)$$

Using Chebyshev's inequality,

$$P(|T_a - \mu| \geq k\sigma) \leq 1/k^2$$

Bounds:

$$T_a \in [-1.3, 1.3]$$



Probabilistic Modeling with Deterministic Decisions

Mathematical equations involving the bounds

CLOUDINESS

$$n \sim N(0.5, 0.2^2)$$

Cloudiness modifies surface radiation through the transfer function:

$$T_{cl} = 1.0 - 0.233n - 0.415n^2$$

Cloudiness is modeled using a normal distribution centered at average sky coverage.



Probabilistic Modeling with Deterministic Decisions

Mathematical equations involving the bounds

ICE ALBEDO

$\alpha \sim \text{Truncated Normal}(0.34, 0.05)$

Bound:

$\alpha \in [0.2, 0.5]$

Ice albedo determines the fraction of solar radiation reflected by the glacier surface.

Deterministic Baseline Logic

- The algorithm always applies the same deterministic rules to compute glacier melt.

Step-by-Step Operation:

1. Sample environmental inputs from probability distributions
2. Compute energy flux using Surface Energy Balance
3. Convert energy to melt depth
4. Aggregate melt values across simulations

Important Constraint:

- No randomized algorithms in the decision logic.
- Only the inputs are stochastic.

Simulation Implementation

Algorithms Used:

- Monte Carlo Simulation
- Deterministic Surface Energy Balance Model

Code Structure Overview:

1. Initialize parameters
2. Generate random samples for environmental variables
3. Apply deterministic melt computation
4. Store results
5. Repeat for large number of iterations

Full Pipeline Execution:

Random Input Generation → Deterministic Processing → Melt Output Collection → Statistical Analysis

Results and Performance Analysis

- **Performance Metrics Used:**

- Mean glacier melt
- Variance of melt estimates
- Confidence intervals
- Probability of extreme melt events

- **Statistical Outputs:**

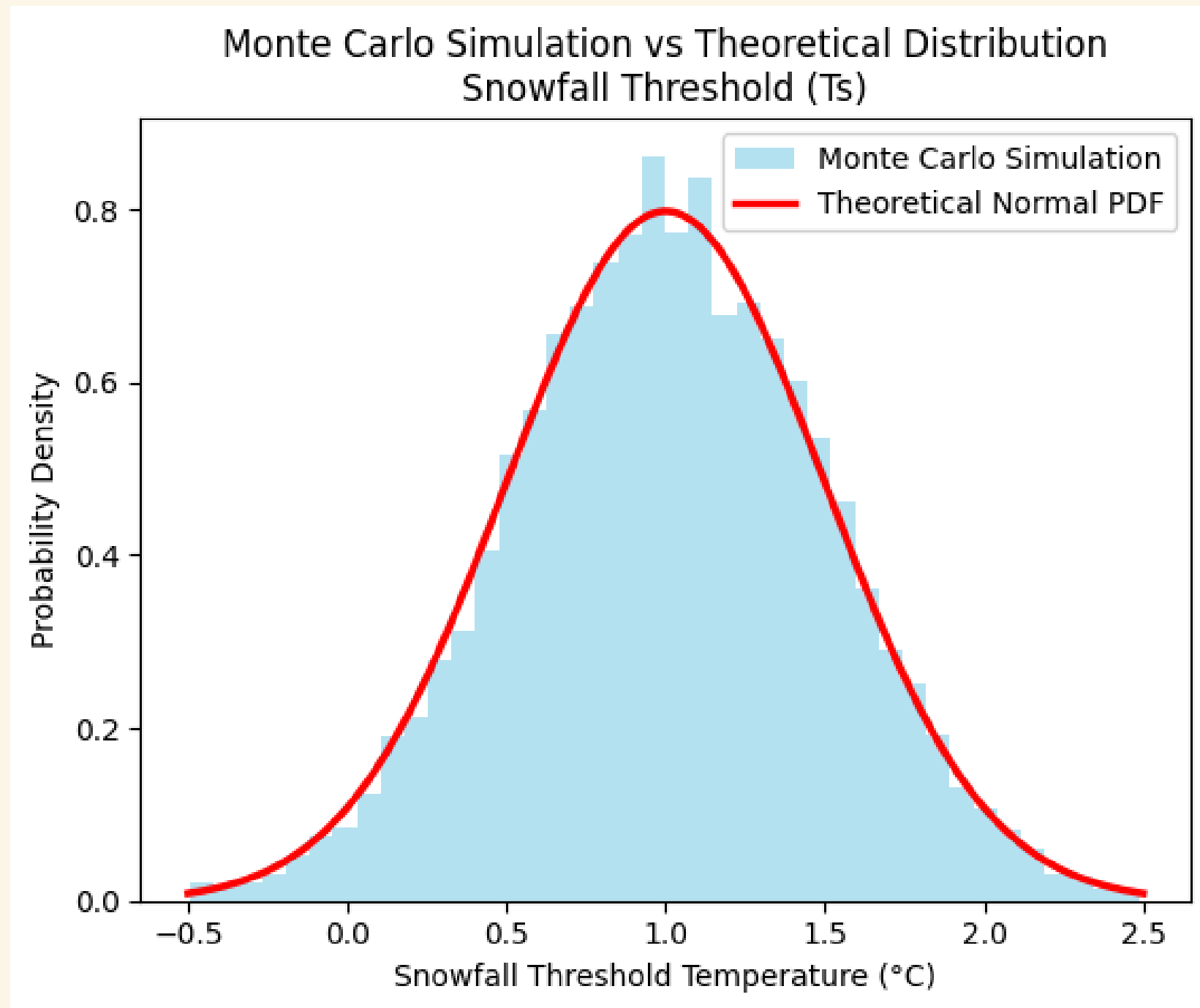
- Probability Density Function (PDF) of melt distribution
- Cumulative Distribution Function (CDF)
- Histogram of simulation outcomes

- **Observed Behavior Under Deterministic Strategy:**

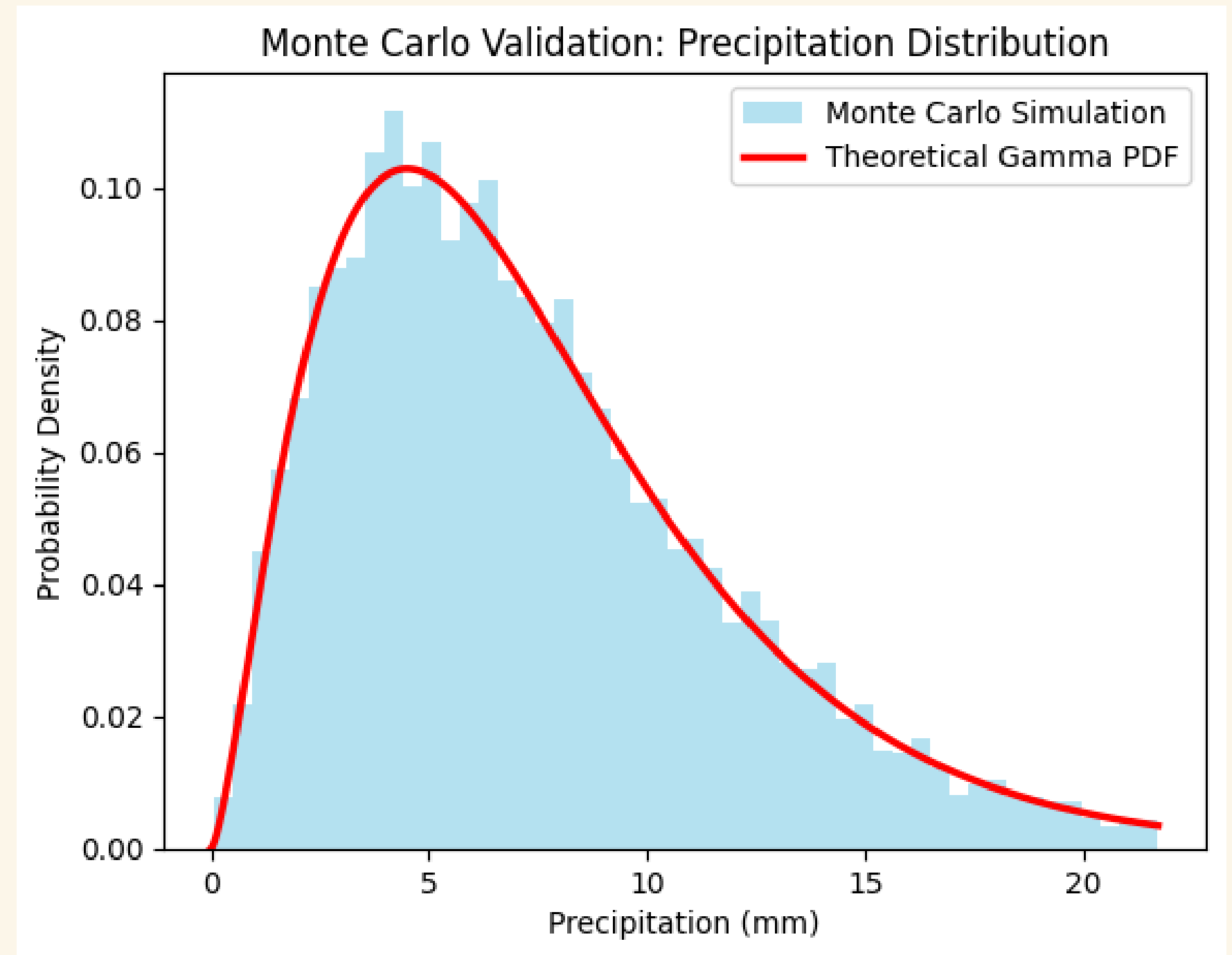
- Stable mean melt estimate as number of simulations increases
- Convergence supported by Law of Large Numbers

Plots of Monte Carlo simulation

Snowfall Threshold

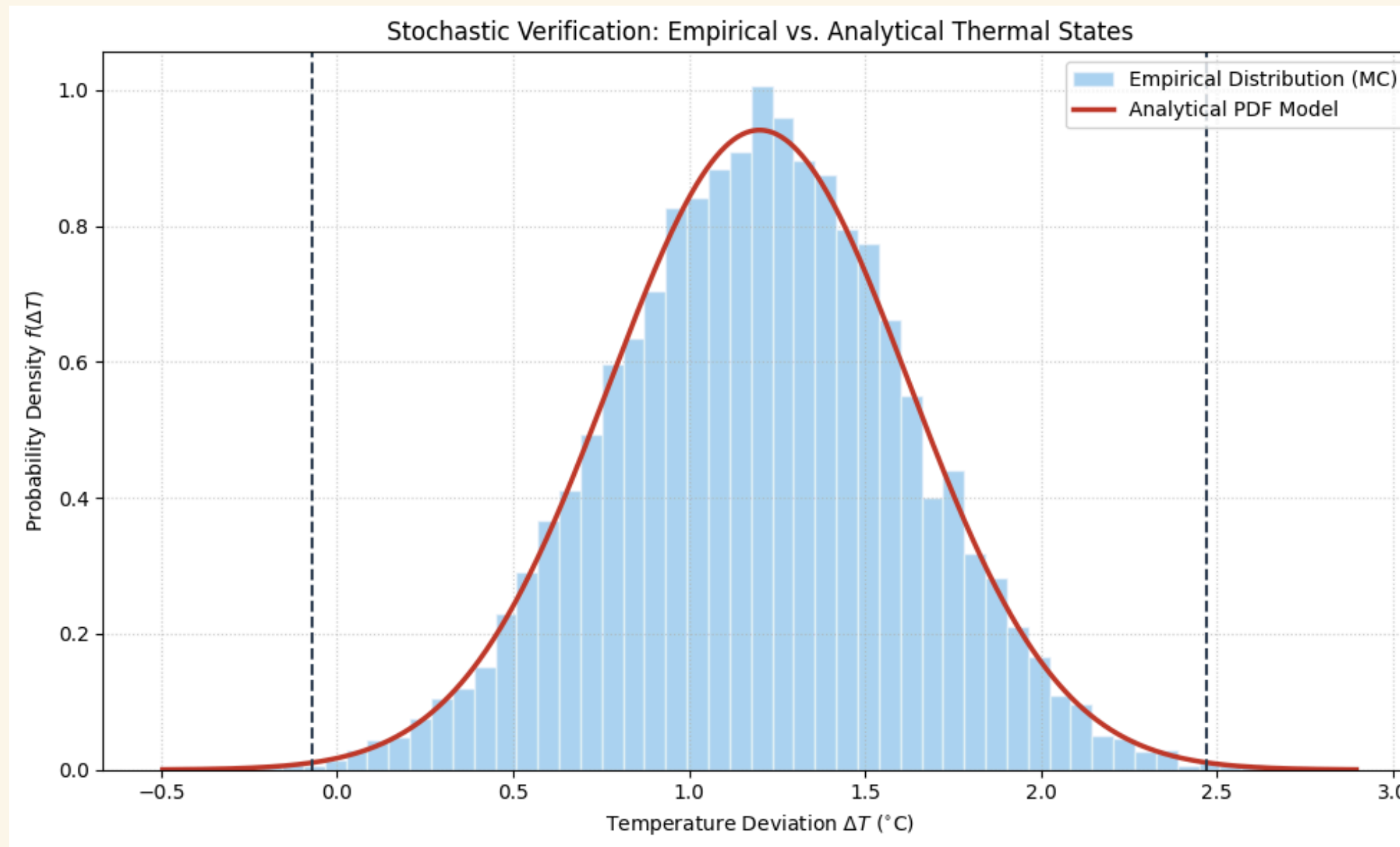


Precipitation

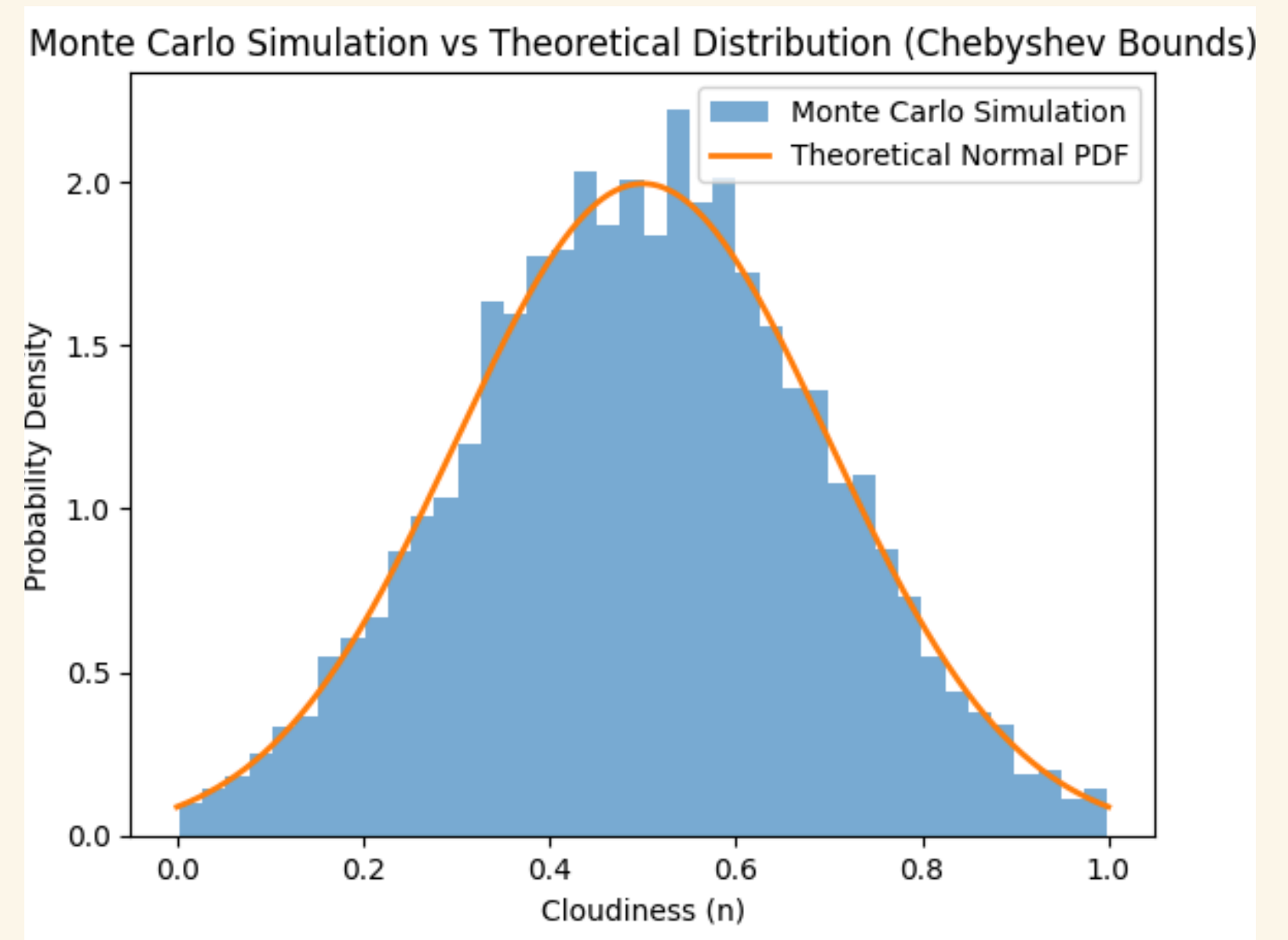


Plots of Monte Carlo simulation

Air Temperature

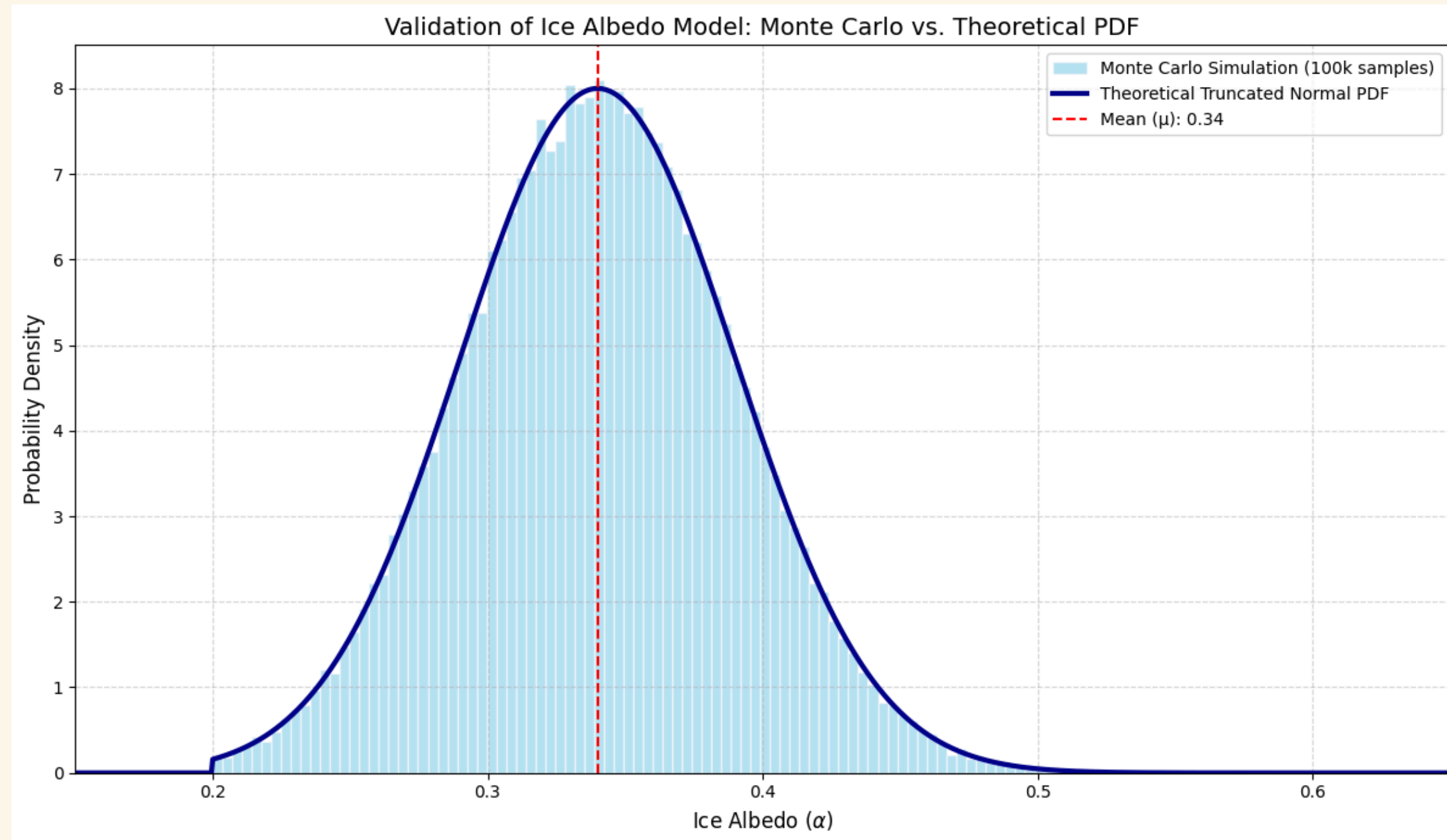


Cloudiness



Plots of Monte Carlo simulation

Ice Albedo



Limitations of Deterministic Strategy

- **Weaknesses of Deterministic Baseline:**

- Cannot adapt decision rules based on changing scenarios
- May underperform under extreme climate variability

- **Simplified Aspects of the Model:**

- Limited environmental variables
- Simplified radiation interactions
- Static decision rules

- **Situations Where Baseline Performs Poorly:**

- Extreme temperature spikes
- Sudden precipitation anomalies
- Non-linear climate feedback effects.

Summary and Transition to Randomized Strategies

- **Key Insights from Deterministic Simulation:**
 - Deterministic models provide a stable baseline.
 - Monte Carlo simulation reveals variability in glacier melt.
- **Why Randomized Approaches May Help:**
 - Allow adaptive responses to uncertainty
 - Improve modeling of complex climate dynamics
- **Planned Refinements for Milestone 4:**
 - Introduce randomized decision mechanisms
 - Improve probabilistic modeling of environmental interactions
 - Explore advanced stochastic optimization strategies.

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Thank You